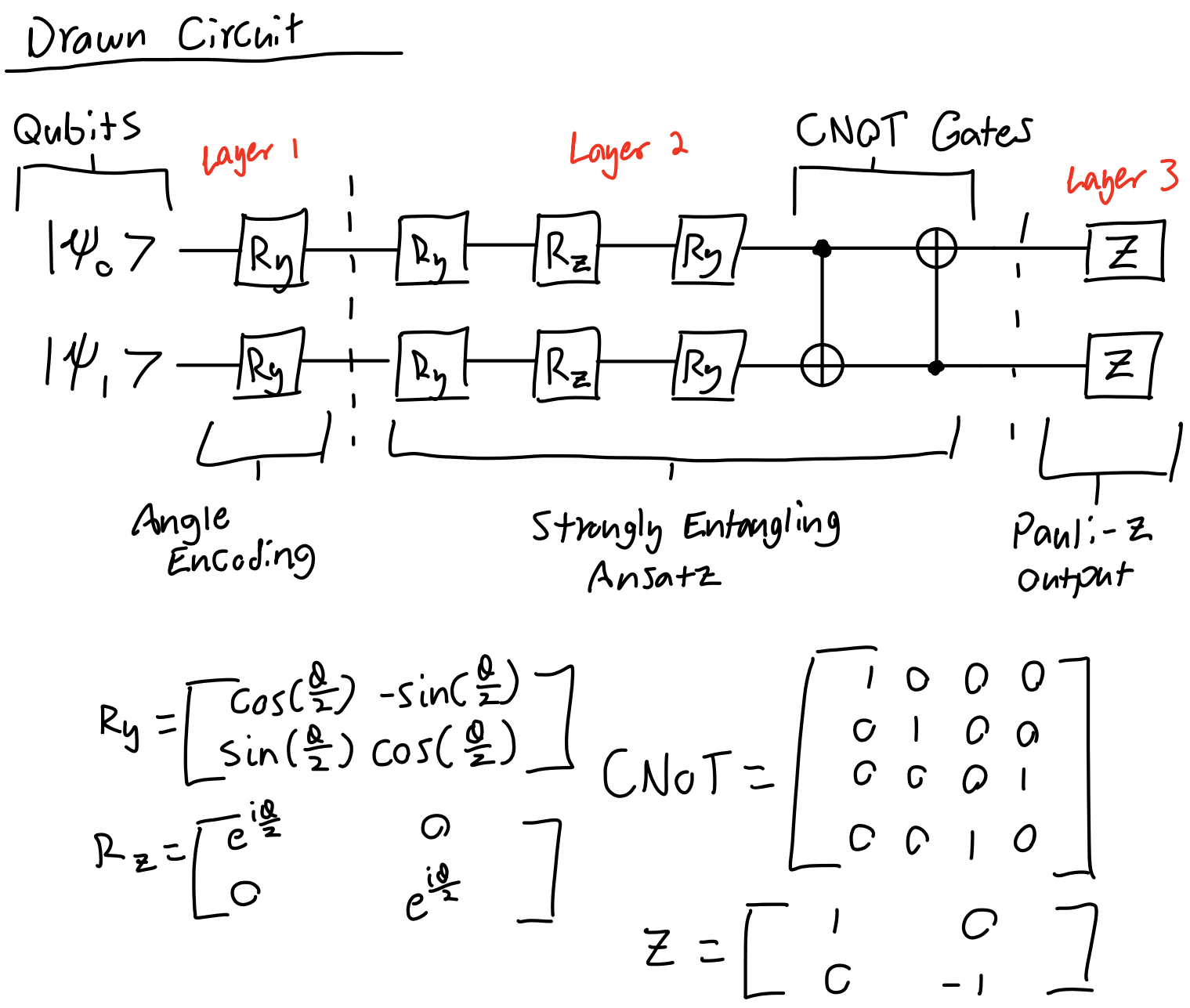
The Two Qubit Circuit

Looking at the output, we can follow one sample in this single layer circuit, where we can go more in depth on the initial encoding, the trainable parameters, the entanglement, and then the final output measurement.

Terminal Output

```
0: ──RY(0.30)──╭RY(0.00)──RZ(0.00)──RY(0.00)─╭●─╭X─┤ <Z>
1: ──RY(-0.40)─╰RY(0.00)──RZ(0.00)──RY(0.00)─╰X─╰●─┤ <Z>
```



$$R_y = \begin{bmatrix} \cos(\frac{\theta}{2}) & -\sin(\frac{\theta}{2}) \\ \sin(\frac{\theta}{2}) & \cos(\frac{\theta}{2}) \end{bmatrix}$$

$$R_z = \begin{bmatrix} e^{i\frac{\theta}{2}} & 0 \\ 0 & e^{i\frac{\theta}{2}} \end{bmatrix}$$

$$CNOT = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Walkthrough

The qubits, represented as $|\psi\rangle = |00\rangle$ (taken from the kronecker product of both qubits: $|\psi_0\rangle \otimes |\psi_1\rangle$), are first fed through the initial angle encoding gates, which in this case are the R_y gates seen at the beginning of the circuit, where we feed our classical inputs into these gates as θ. These initial gates do not contain trainable parameters and are strictly for allowing us to introduce the classical inputs into our quantum circuit.

The trainable parameters come in the next step where we enter the strongly entangling ansatz, containing the gates Ry, Rz, and Ry as seen in the circuit sketch, which rotate each qubit. These each contain a trainable parameter, and across both qubits there are a total of 6 trainable angle parameters the model can learn and adjust.

The next step is entanglement using the CNOT gates, going from wire 0 -> 1, then from wire 1-> 0. The CNOT gates do not have trainable parameters and are meant for entanglement.

The final gates are the Pauli-Z expectation value gates,$\langle Z_0 \rangle$ and $\langle Z_1 \rangle$, and returns two numbers, one output per qubit. The circuit returns two numbers because it measures one Pauli-Z expectation per wire. The classical readout learns how to weight those two values to produce a single classification score.

Taking a look at how the shapes look for this circuit, the overall weight tensor shape should be (1,2,3), where there is 1 layer, 2 qubits, and three trainable angle parameters per qubit. The input should be a shape of (2,) or $[x_0, x_1]$ and the output should be a shape of (2,) or $[z_0, z_1]$